

DP-aware MIA (target $\epsilon = 0.5$)

shadows: 8 DP-SGD EEGNets

target: 1 DP-SGD EEGNet, $\epsilon_{\text{final}} = 0.498$

members vs non-members: 52 vs 52

advantage (TPR-FPR) = 0.038

MIA AUC = 0.449
unfounded EEGNet baseline AUC: 0.478
95% CI: [0.334, 0.558]

DP-aware shadow MIA against DP-SGD $\epsilon = 0.5$